

Lab Sheet 10

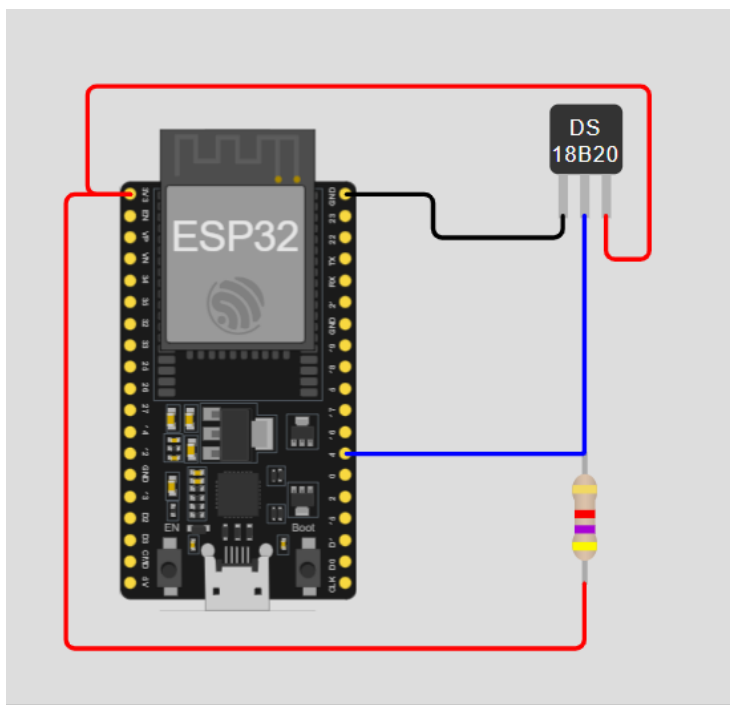
Title: IoT Sensor Data Logging System using ESP32 and Google Sheets.

Aims:

- To design and implement an IoT-based data logging system using an ESP32 microcontroller and a temperature sensor, where real-time sensor data is transmitted over Wi-Fi and stored in a Google Sheets spreadsheet. This system demonstrates cloud-based data storage, real-time monitoring, and basic IoT integration using HTTP requests and Google Apps Script.

Task 01: IoT Sensor Data Logging System Setup

1. Project Setup in Wokwi / VS Code
 - a. Name the project **SensorData_ESP32_GoogleSheet**.
2. Circuit Design
 - a. Sensor Connection
 - i. Connect the DS18B20 (One-Wire Digital Temperature Sensor) to the ESP32.

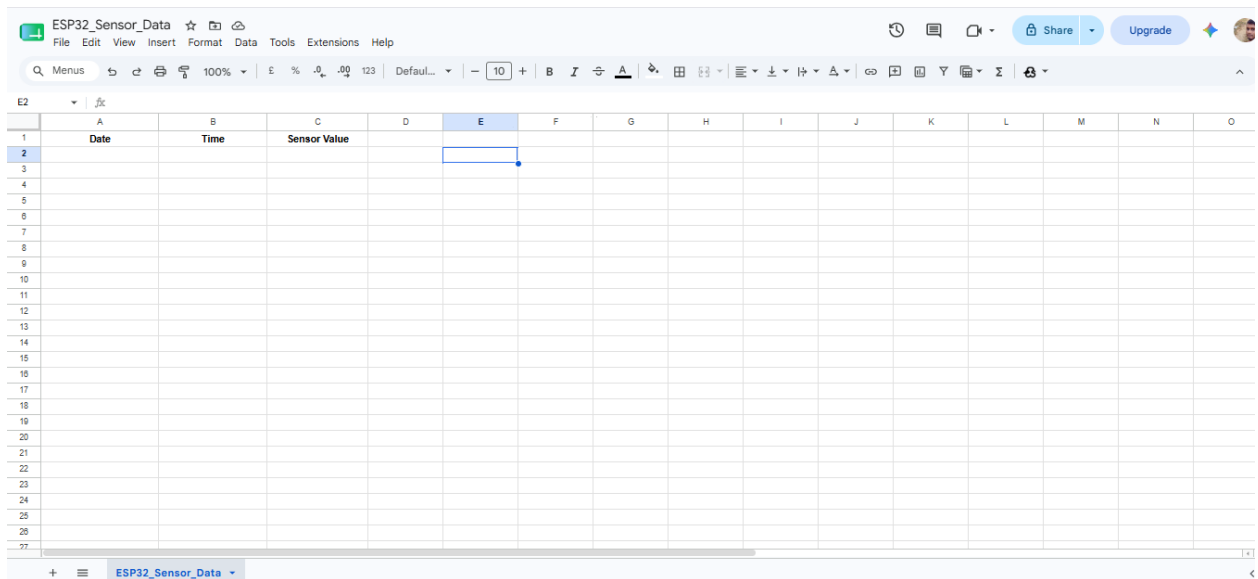


3. Library Integration

- WiFi (Built-in ESP32 library)
- HTTPClient (Built-in ESP32 library)
- OneWire (External library)
- DallasTemperature (External library)

4. Google Sheets Configuration

- a. Create a new Google Sheet
- b. Name the sheet **ESP32_Sensor_Data**
- c. Create the following column headers:
 - i. Date
 - ii. Time
 - iii. Sensor Value



- d. Open Google Apps Script from the Extensions menu
- e. Write a script to receive HTTP GET/POST requests from ESP32

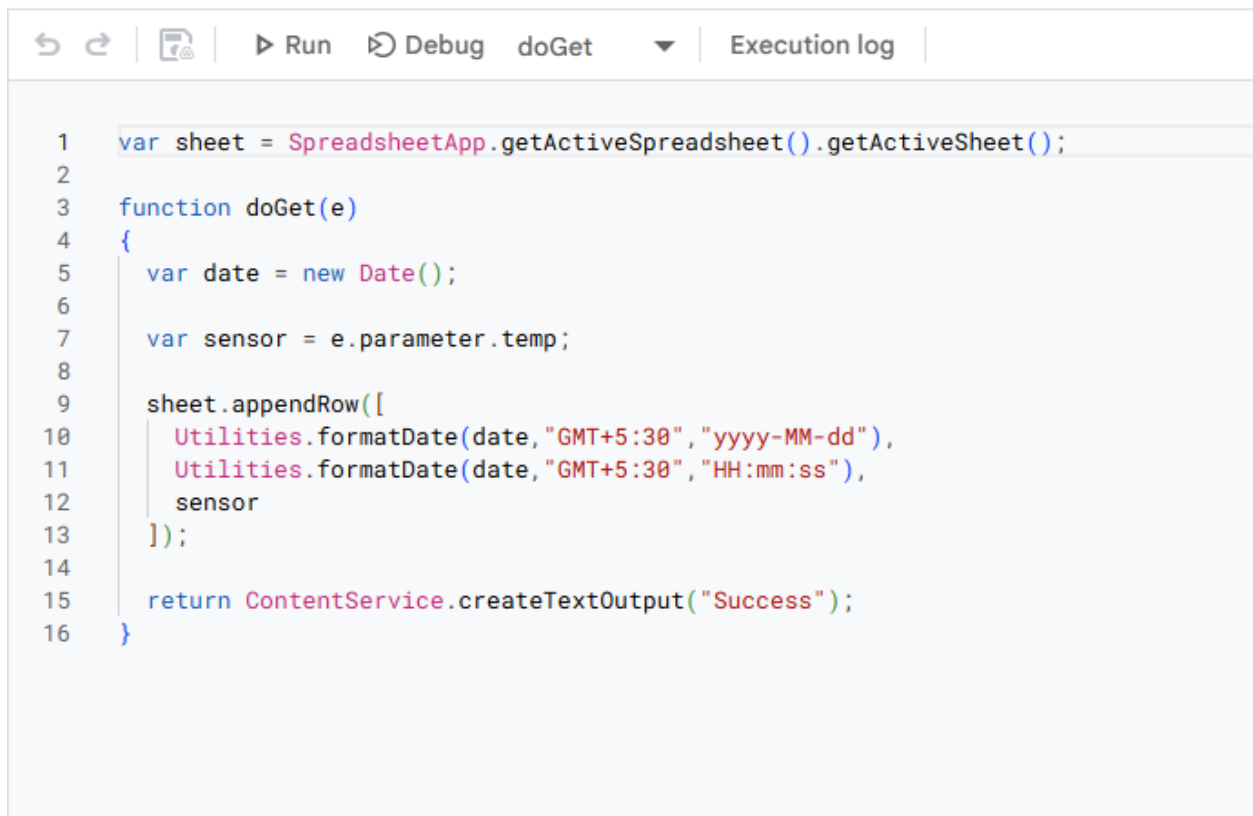
```
var sheet = SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();

function doGet(e)
{
    var date = new Date();

    var sensor = e.parameter.temp;

    sheet.appendRow([
        Utilities.formatDate(date, "GMT+5:30", "yyyy-MM-dd"),
        Utilities.formatDate(date, "GMT+5:30", "HH:mm:ss"),
        sensor
    ]);

    return ContentService.createTextOutput("Success");
}
```



```
1  var sheet = SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();
2
3  function doGet(e)
4  {
5      var date = new Date();
6
7      var sensor = e.parameter.temp;
8
9      sheet.appendRow([
10         Utilities.formatDate(date, "GMT+5:30", "yyyy-MM-dd"),
11         Utilities.formatDate(date, "GMT+5:30", "HH:mm:ss"),
12         sensor
13     ]);
14
15     return ContentService.createTextOutput("Success");
16 }
```

- f. Deploy the script as a Web App

Manage deployments

Active	Configuration
Untitled	Version Version 1 on 5 Jul 2026, 14:21 Description Deployment ID AKfycbwHEwHLWSVRaAjUB69jsKUGvT_aW2AcFX_MicxC9c9yrRh5XBHwBtY4Y-JMR... Copy Web app URL https://script.google.com/macros/s/AKfycbwHEwHLWSVRaAjUB69jsKUGvT_aW2Ac... Copy Execute as
Archived	
No archived deployments	

[Cancel](#) [Deploy](#)

- g. Set access permission to Anyone

- h. Copy the generated Web App URL

https://script.google.com/macros/s/AKfycbwHEwHLWSVRaAjUB69jsKUGvT_aW2AcFX_MicxC9c9yrRh5XBHwBtY4Y-JMR38Uok4r/exec

5. Programming Logic Development in Wokwi / VS Code

- Initialize Serial Monitor
- Initialize Wi-Fi connection
- Initialize the DS18B20 sensor
- Read temperature values at regular intervals
- Send sensor data to Google Sheets using an HTTP requestData should be sent automatically at a fixed time interval (e.g., every 2 seconds)
 - Each sensor reading should be stored as a new row in Google Sheets

Code:

```
#include <Arduino.h>
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <HTTPClient.h>
#include <OneWire.h>
#include <DallasTemperature.h>

const char* ssid = "Wokwi-GUEST";
const char* password = "";

String scriptURL =
"https://script.google.com/macros/s/AKfycbwHEwHLWSVRaAjUB69jsKUGvT_aW2AcFX_MicxC9
c9yrRh5XBHwBtY4Y-JMR38Uok4r/exec";

// DS18B20 Data Pin
#define ONE_WIRE_BUS 4

OneWire oneWire(ONE_WIRE_BUS);

DallasTemperature sensors(&oneWire);

void setup()
{
  Serial.begin(115200);

  sensors.begin();

  WiFi.begin(ssid, password);

  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi Connected");
}

void loop()
{
```

```

sensors.requestTemperatures();

float temp = sensors.getTempCByIndex(0);

Serial.print("Temperature : ");
Serial.print(temp);
Serial.println(" °C");

if (WiFi.status() == WL_CONNECTED)
{
    WiFiClientSecure client;
    client.setInsecure();

    HTTPClient http;

    String url = scriptURL + "?temp=" + String(temp);

    http.setFollowRedirects(HTTPC_FORCE_FOLLOW_REDIRECTS);
    http.begin(client, url);

    int httpCode = http.GET();

    Serial.print("HTTP Response : ");
    Serial.println(httpCode);

    http.end();
}

delay(2000);
}

```

6. Simulation & Testing in Wokwi / VS Code

- a. Simulate the circuit in Wokwi
- b. Verify temperature readings in the Serial Monitor
- c. Confirm real-time data entries in Google Sheets